

STEMBuddies Canada

**Sex-Based Differences in Cognitive and Pathological Response to E2814: An Anti-Tau
Therapy for Early-Stage Alzheimer's Disease**

Amodita Arora, Arya Devanath, Leslie Gao

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Abstract

Alzheimer's disease (AD) is a neurodegenerative disorder characterized by cognitive decline from accumulation of the microtubule-associated protein tau. E2814 is an anti-tau monoclonal antibody that inhibits extracellular tau progression. While evidence highlights sex-based differences in tau pathology, most therapeutic trials overlook sex as a biological variable. This study aims to evaluate how tau pathology and cognitive response differ between male and female patients aged 65+ with early-stage AD treated with E2814. In a 24-week randomized, double-blind, placebo-controlled trial, 200 participants received E2814 or placebo. Cognition was measured using ADAS-Cog13, and tau burden was assessed via PET imaging at baseline, 12, and 24 weeks. The sample size of 200 was based on prior studies of E2814 and was powered to detect moderate treatment-by-sex interaction effect sizes with 80% power at $\alpha = 0.05$. Two-way ANOVA indicated that females receiving E2814 improved cognitively more than males and placebo groups. Mixed-effects models confirmed sex and treatment as significant predictors of cognitive outcome. Correlation analyses revealed a stronger link between tau reduction and cognitive improvement in women.

Keywords

Alzheimer's disease, tau pathology, E2814, monoclonal antibody, sex differences, PET imaging, cognitive decline, ADAS-Cog13, immunotherapy, neurodegeneration

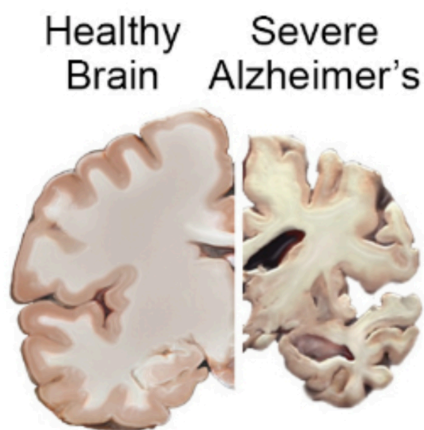
Introduction

As of 2025, AD causes one in three deaths, exceeding the combined mortality rates of breast and prostate cancer (Alzheimer's Association, 2025). It is a progressive genetic and protein-based

neurodegenerative disorder caused by the buildup of neurofibrillary tangles of hyperphosphorylated tau (Pallawi & Singh, 2023). This accumulation impairs synaptic communication, triggers neuroinflammation, and cognitive decline (see Figure 1). By causing memory loss, disorientation, language and movement difficulties, behavioral changes, it progresses to complete dependence, and ultimately, death. E2814 is a Phase II/III anti-tau monoclonal IgG1 antibody targeting the microtubule-binding region of tau to prevent the spread of tau (See Figure 2) (Hanger, & Noble, 2019; McLoughlin, 2022; Wu et al., 2021). While early results are promising, sex-specific cognitive responses to E2814 remain unexamined. This oversight is critical given that sex influences disease progression, pathology, and treatment response (Canevelli et al., 2017). With implications for personalized, sex-specific approaches to treatment, this study investigates whether E2814 produces different cognitive outcomes in males and females with early-stage AD.

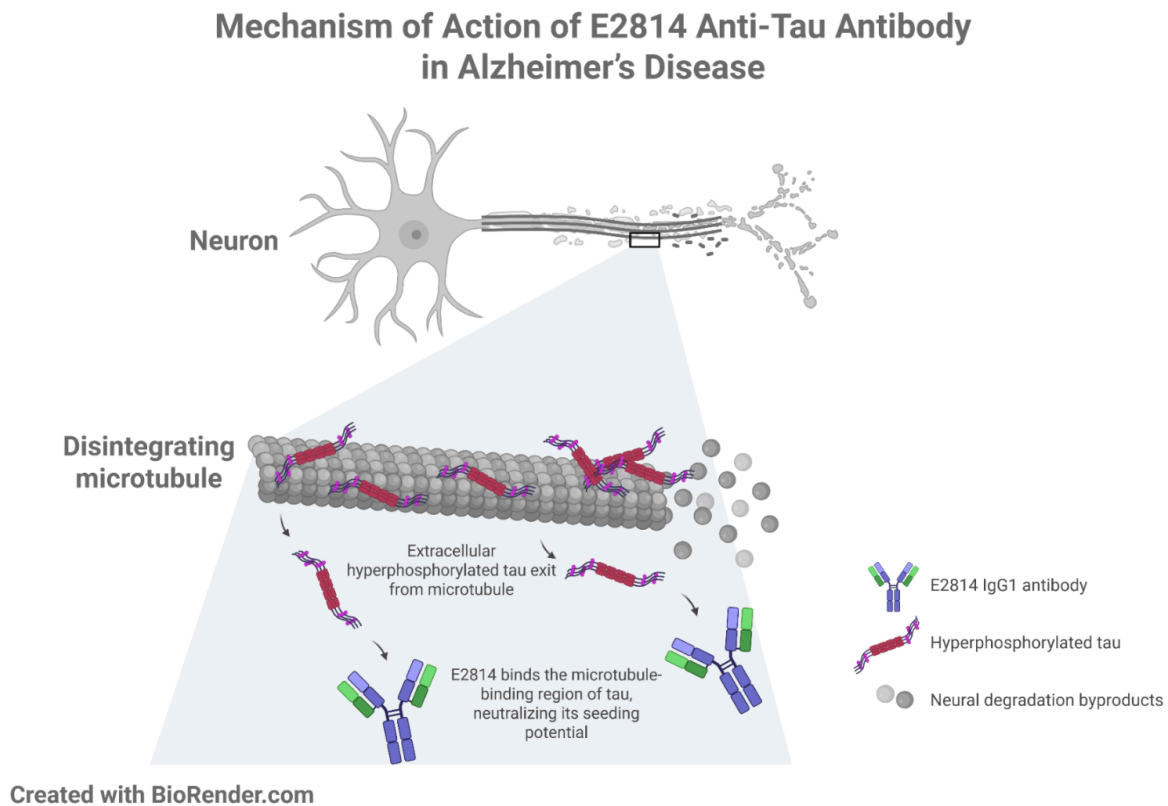
Figure 1

Healthy vs. Diseased Brain



Note. Atrophy present in the cerebral cortex and hippocampus.
(Pallawi & Singh 2023)

Figure 2
Mechanism of Action



Note. Hyperphosphorylated tau proteins detach from microtubules, exiting into extracellular space where they are bound by E2814. E2814 neutralizes tau's seeding ability, preventing further trans-synaptic tau spread.

(Goodwin et al., 2020; Hanger, & Noble, 2019; Lista et al., 2024; McLoughlin, 2022; Song et al., 2023; Wu et al., 2021)

Hypothesis

Given that biological sex influences tau pathology in AD, male and female patients aged 65+ with early-stage AD will exhibit distinct cognitive responses and tau accumulation to E2814 compared to placebo. Women are expected to show improved cognitive response due to higher baseline tau burden.

Rationale

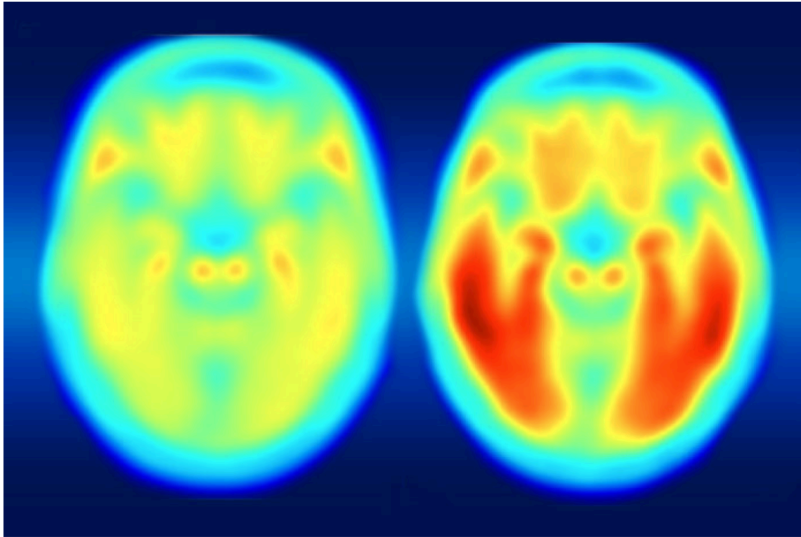
In individuals aged 65+, research suggests that biological sex shapes tau pathology. Women carrying the APOE4 allele exhibit faster tau accumulation and higher levels of cerebrospinal phosphorylated tau than men (George, 2025; Smith et al., 2020). Despite these findings, anti-tau antibody trials, including those investigating E2814, do not report sex-stratified outcomes (Smith, 2023). This omission limits our understanding of how biological differences between males and females drive divergent cognitive responses to treatment. This research could improve therapeutic efficiency, reduce trial failures, and guide precision medicine.

Methodology

This study uses a randomized, double-blind, placebo-controlled design to investigate how biological sex influences cognitive responses to anti-tau antibody treatment. Approximately 200 individuals (100 men, 100 women) with early-stage AD (MMSE scores 20-26), aged ≥ 65 , with stable medications for ≥ 3 months. Exclusion criteria include other dementias, psychiatric illness, traumatic brain injury, or medical conditions affecting cognition or drug metabolism. Participants are randomly assigned to receive either E2814 or a visually identical placebo intravenously, dosed per Phase II protocols (Eisai Inc, 2025). Randomization is stratified by sex, age, and APOE4 status. The placebo will be a sterile saline solution administered the same way as E2814. Cognitive performance will be assessed using the ADAS-Cog13 at baseline, 12 weeks, and 24 weeks (Skinner et al., 2012). Researchers will measure tau pathology by PET imaging using standardized uptake value ratios (SUVRs) at baseline and 24 weeks (Strait, 2016) (See Figure 3). Longitudinal data will track sex-specific patterns in cognitive and pathological changes.

Figure 3

PET Image of a Healthy vs. Diseased Brain



Note. Elevated tau accumulation shown in red/yellow (Strait, 2016)

A two-way ANOVA will evaluate the main effects of treatment (E2814 vs. placebo), sex (male vs. female), and their interaction on cognitive and pathological outcomes. Repeated measures ANOVA will assess changes over time. Mixed-effects linear regression will adjust for confounders including APOE4 status, hormone levels, and social factors. Pearson correlation will explore within-sex associations between tau changes and cognitive performance. All participants will provide written consent, and the protocol will receive IRB approval and be registered on ClinicalTrials.gov. Data handling will follow ethical guidelines.

Expected results

Based on current literature and an interview with RN Mary Zheng (Hennick Bridgepoint Hospital), the 24-week trial is expected to show significant sex-based differences in response to

E2814. Two-way ANOVA will reveal that females receiving E2814 demonstrate greater cognitive improvement on ADAS-Cog13 than males or placebo groups. PET imaging will show a greater reduction in tau among treated females, with men showing smaller changes and placebo groups remaining stable (Meglio, 2024). Mixed-effects models will identify sex and treatment as key predictors of outcomes. Pearson correlations suggest a stronger link between tau reduction and cognitive gains in women (Nebel et al., 2023). Overall, findings support the need for sex-specific approaches in future AD trials.

Conclusion

This study examines how biological sex influences tau pathology and cognitive outcomes in AD treatment, a factor often overlooked in clinical research. Our expected results suggest that women may benefit more from E2814 than men due to higher baseline tau burden and distinct clearance dynamics, highlighting the urgency of stratifying clinical trials by sex for improved precision. Future research should validate these results and investigate the molecular mechanisms behind sex-based differences in tau pathology. These insights might shape prospective FDA regulations for AD trials, foster incorporation of sex-stratified reporting in therapeutic studies, and inform individualized intervention strategies in clinical neurology.

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Definition Sheet

1. ADAS-Cog13 (Alzheimer's Disease Assessment Scale – Cognitive, 13-item version): A commonly used cognitive test in AD research that assesses memory, reasoning, orientation, and language skills.
2. Alzheimer's Disease (AD): A progressive brain disorder that causes memory loss, cognitive decline, and behavioral changes. It is marked by the buildup of amyloid plaques and tau tangles.
3. APOE4 (Apolipoprotein E ϵ 4 allele): A genetic variant associated with increased risk for Alzheimer's disease. People with this allele often show faster disease progression and more tau buildup.
4. Baseline: The starting point of data collection in a study before any treatment is given to compare against future measurements.
5. Cognitive Decline: The gradual worsening of thinking, memory, decision-making, and reasoning abilities, common in aging and neurodegenerative diseases like AD.
6. Confounding Variable: A variable that might influence both the treatment and outcome, potentially skewing results if not controlled. Examples: age, genetics, hormone levels.
7. Correlation Analysis: A way to determine whether two variables are related, such as tau levels and memory test scores.
8. E2814: An experimental (Phase II/III) anti-tau monoclonal IgG1 antibody that targets and binds to abnormal tau protein in an effort to slow its spread between brain cells.
9. Intravenous (IV) Administration: A method of delivering treatment directly into a patient's vein using a needle or tube.

10. Mini-Mental State Examination (MMSE): A 30-point test used to screen for cognitive impairment. Scores between 20–26 indicate early-stage Alzheimer's.
11. Mixed-Effects Linear Regression Model: A statistical method that considers both fixed factors (like sex and treatment group) and random factors (like individual differences), especially useful for long-term or repeated measurements.
12. Monoclonal Antibody (mAb): Lab-made proteins that bind specifically to other proteins. In this case, E2814 targets tau protein to help stop its spread in the brain.
13. Neurofibrillary Tangles: Clumps of twisted tau proteins found in brain cells of individuals with AD, interfering with communication between neurons and contributing to cognitive decline.
14. Pearson Correlation Coefficient (r): A measure of the strength and direction of a relationship between two variables. In this study, it's used to explore whether changes in tau are linked to cognitive improvement.
15. PET Imaging (Positron Emission Tomography): A brain imaging technique that uses radioactive tracers to detect biological changes, like tau accumulation, in real time.
16. Placebo: A harmless, inactive substance designed to look like the real treatment, used as a control in studies to compare effectiveness.
17. Randomized, Double-Blind, Placebo-Controlled Trial: A gold-standard clinical trial design where participants are randomly assigned to treatment or placebo, and neither researchers nor participants know who receives which. This reduces bias and strengthens the validity of findings.
18. Repeated Measures ANOVA: A statistical test used to measure changes in the same participants over time, such as before and after treatment.

19. Sex as a Biological Variable (SABV): A research principle that recognizes biological sex can influence health, disease progression, and treatment response. Studies that account for SABV analyze men and women separately.
20. Tau Burden: The amount or concentration of abnormal tau protein in the brain, which can be measured through imaging or fluid biomarkers.
21. Tau Clearance Dynamics: The rate and efficiency at which abnormal tau is removed from the brain. This may differ by sex and can affect treatment success.
22. Tau Protein: A protein that normally stabilizes internal support structures in brain cells. In AD, tau becomes defective and forms tangles, disrupting brain function.
23. Two-Way ANOVA (Analysis of Variance): A statistical test that analyzes how two independent variables (e.g., sex and treatment) affect an outcome (e.g., cognitive performance).

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